



SESSION 11

Chi-Square Tests

Chi-Square Distribution · Goodness-of-Fit · Test of Independence ·
Contingency Tables

Chi-Square Distribution Overview

Chi-Square Goodness-of-Fit Test

Chi-Square Test of Independence

Contingency Tables & Expected Frequencies

Degrees of Freedom Calculations

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Python Code Implementation

Session Notes — Complete Reference

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01 • Chi-Square Distribution Overview

The **Chi-Square (χ^2) distribution** is a continuous distribution of the sum of squared standard normal variables. It is right-skewed and defined only for positive values, parameterized by its degrees of freedom (k).

02 · Chi-Square Goodness-of-Fit Test

Determines if a sample matches an expected categorical distribution (e.g., testing if a die is fair).

GOODNESS-OF-FIT FORMULA

$\chi^2 = \sum [(O_i - E_i)^2 / E_i]$ where O_i = Observed frequency, E_i = Expected frequency

03 · Chi-Square Test of Independence

Used to determine whether two categorical variables are independent or associated (e.g., testing if gender and political affiliation are related).

04 · Contingency Tables & Expected Frequencies

Observations are structured in a matrix. Under the null hypothesis of independence, expected frequency $E_{i,j}$ for each cell is calculated as:

EXPECTED FREQUENCY FORMULA

$$E_{(i,j)} = (\text{Row Total} \cdot \text{Column Total}) / \text{Grand Total}$$

05 • Degrees of Freedom Calculations

- **Goodness-of-Fit:** $df = c - 1$ (where c is the number of categories).
- **Test of Independence:** $df = (r - 1) \times (c - 1)$ (where r is rows and c is columns).

06 • Assumptions of Chi-Square Tests

For valid Chi-Square results:

1. **Random sampling:** Randomly selected data.
2. **Mutually exclusive categories:** Each case fits into exactly one category.
3. **Sample size:** Expected frequency in each cell must be ≥ 5 .

07 · Python Code Implementation

```
import numpy as np
from scipy import stats

# Contingency table (Rows: Gender, Columns: Product Choice)
observed = np.array([[50, 30, 20],
                     [45, 35, 20]])

# Perform Chi-Square Test of Independence
chi2, p_val, df, expected = stats.chi2_contingency(observed)
print(f"Chi2: {chi2:.4f}, p-value: {p_val:.4f}, df: {df}")
```

08 · Quick Reference Card

Chi-Square Test Quick Reference

- **Goodness-of-Fit:** Compares observed vs expected in one variable.
- **Independence:** Tests association between two categorical variables.
- **Expected Cell Threshold:** Ensure expected frequencies ≥ 5 .